LESSON 10.1

Math is Everywhere

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find applications of math outside of school — in professions like being a chef, in hobbies like gardening, and around my own house.

WHY THIS MATTERS Some people think math is just a school subject with little relevance outside the classroom. But math is everywhere — chefs use it all day in restaurant kitchens, gardeners use it in planter beds, and it is hiding in jobs and hobbies all around you. A chef scales a recipe up for forty covers and works out how much stock to order. A gardener measures a planter bed and calculates the soil needed to fill it. Today you will name the math hiding inside jobs like these, and you will do some of it yourself — area, volume, rates, and estimates that have to be close enough to act on.

REMEMBER

Math is Everywhere

Mathematical Modeling in Careers & Daily Life Chefs supervise a team of cooks preparing everything from appetizers to desserts. Watch how much of that job is quietly mathematical.

1. Frame real-world problems mathematically using rates, percentages, and geometry.
2. Collect and analyze data to optimize decisions and resource allocation.
3. Communicate quantitative findings clearly with equations and visual models

Words: profession *(profesión)* · inventory *(inventario)* · portion *(porción)* · area *(área)* · volume *(volumen)*

PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

Sort each gardening question: does it need area or volume?

Groups: Area (square feet) · Volume (cubic feet)

_____ How much ground does the planter cover?

_____ How much mesh covers the top against birds?

_____ How much of the yard is left uncovered?

_____ How much soil fills the planter?

_____ How much compost fills a second, deeper planter?

_____ How much water does the full rain barrel hold?

HINT Covering flat ground is area; filling a container is volume.

2 **Circle the letter of the best answer. Show how you know.**

Which statement matches this lesson's big idea?

(A) Math is just a school subject with little relevance outside the classroom

(B) Only professions like chef use math; hobbies do not

(C) Math is everywhere — in professions, hobbies, and problems around your house

(D) Math is only useful for people who work with money

HINT Think about all three contexts in this lesson.

SHOW YOUR WORK

3 **Complete the table. Show how you found each value.**

Each person who rinses instead of running water saves 8 gallons a day. Complete the table.

People in the house	Thinking	Gallons saved per day
3		
5		
8		

HINT Each person contributes the same 8 gallons — equal groups multiply.

SHOW YOUR WORK

PART 2 Practice the skill

show every step

4 Circle the letter of the best answer. Show how you know.

What is 0.3 × 5?

(A) 0.15
(B) 1.5
(C) 5.3
(D) 15

SHOW EACH PARTIAL PRODUCT, THEN ADD THEM

1. **Split** *Break one factor apart.*
2. **Multiply** *One part at a time.*
3. **Add** *Add the parts.*

5 Circle the letter of the best answer. Show how you know.

What is 0.4 × 0.7?

(A) 0.028
(B) 0.28
(C) 2.8
(D) 28

SHOW EACH PARTIAL PRODUCT, THEN ADD THEM

1. **Split** *Break one factor apart.*
2. **Multiply** *One part at a time.*
3. **Add** *Add the parts.*

PART 3 Apply it

real-world problems

6 **Circle the letter of the best answer. Show how you know.**

A notebook costs $2.45. How much do 6 notebooks cost?

(A) $1.47

(B) $14.10

(C) $14.70

(D) $147.00

SHOW YOUR WORK

EXPLAIN YOUR THINKING

Pick a card you and your partner assigned to different workers. Which measurement does that job actually need — area, volume, or a portion — and how do you know from the task itself?

SENTENCE STARTER The ___ needs ___ because the job is about ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

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PART 1 Understand the idea

models and meaning

1 **Circle the letter of the best answer. Show how you know.**

Which statement best shows what this lesson means by 'math is everywhere'?

(A) Math only appears in jobs that involve money

(B) Math is a school subject with little relevance to life outside the classroom

(C) Only trained professionals like chefs are able to use math

(D) People use math every day in their professions and hobbies, and it solves problems around the house too

SHOW YOUR WORK

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

In 1.2 × 0.35, how many decimal places does the product have?

(A) 0
(B) 1
(C) 2
(D) 3

SHOW EACH PARTIAL PRODUCT, THEN ADD THEM

3 **Circle the letter of the best answer. Show how you know.**

Math is an integral part of nearly every profession, and many hobbies include applications of math. You just saw it in a restaurant kitchen and a vegetable garden. A chef is deciding how many portions of a new dish to prepare tonight, based on how many customers are expected. Which math task is this?

(A) Projecting how much food is needed for the day
(B) Setting the price of the dish
(C) Decorating the plate for presentation
(D) Naming the dish on the menu

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Math is an integral part of nearly every profession, and many hobbies include applications of math. You just saw it in a restaurant kitchen and a vegetable garden. A gardener knows the size of their planter box and wants to know how much soil to buy to fill it. What should the gardener calculate?

(A) The area of the planter
(B) The perimeter of the planter
(C) The number of leaves per plant
(D) The volume of the planter

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

Math is an integral part of nearly every profession, and many hobbies include applications of math. You just saw it in a restaurant kitchen and a vegetable garden. Leaving the water running while brushing wastes an average of 4 gallons compared to turning the water on only to rinse. If one person brushes twice a day for a year (365 days), how many gallons would they save by turning the water off?

(A) 730 gallons

(B) 1,460 gallons

(C) 2,920 gallons

(D) 5,840 gallons

SHOW YOUR WORK

EXPLAIN YOUR THINKING

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